The J-space in OLMo-3-32B-Think:
an open-weights replication of "Verbalizable Representations Form a Global Workspace in Language Models"

special-lab-1, interpretability labs

July 26, 2026

run `2026-07-25_1726` · lens `jlens` @ `581d3986` · seeds 0

v2 delta run in progress (§9): results below marked [v2] update as instruments land

Abstract

We replicate the core claims of Anthropic's July 2026 "global workspace" paper on **Olmo-3-32B-Think**, a fully open (weights + training data) reasoning model, using the reference Jacobian-lens implementation. The *geometry* replicates: a sparse, verbalizable subspace concentrated in the middle layers, whose directions are read by 73–94 downstream components (vs. 0 for random directions). Its *capacity* is roughly an order of magnitude thinner than reported for Claude (≤0.67% of activation variance vs. 6–10%; ∼6 active concepts vs. ∼25). The paper's headline *causal dissociation does not replicate* at our doses: static J-space ablation is indistinguishable from random controls, while matched-dimension *non-verbalizable* high-variance directions cause the real selective damage. Exploiting the thinking variant, we add a result the paper could not test: the workspace does *not* contain the answer before reasoning begins (median rank 5613), but **leads the chain of thought by a median 46 tokens** during reasoning, and is loaded on demand when thinking is suppressed (rank 211 at the answer position, collapsing monotonically L30→L60). An eval-awareness probe finds a highly consistent "being-evaluated" readout direction ($p < 10^{-3}$ on all probed layers) that is semantically legible yet behaviorally inert under ablation.

1 Setup

Model. `allenai/Olmo-3-32B-Think` (64 layers, $d = 5120$), bf16 on one 96 GB RTX PRO 6000 Blackwell — full-precision tier: no quantization, no gradient checkpointing. **Lens.** Reference `jlens` package (`anthropics/jacobian-lens`), fit on 120 WikiText-103 prompts × 128 tokens (the neuronpedia reference recipe; `jlens` README reports saturation near 100; the paper used 1000). Source layers: dense middle band L20–44 (step 2) + early/late controls {4,8,12,16,48,52,56,60}; target L63; four 30-prompt slices merged via `JacobianLens.merge`. Fit cost 5.2 h (∼157 s/prompt, peak 80.6 GB); per-prompt max $\|J\|/\sqrt{d}$ stable at 0.94–0.95 and running-mean deltas fell $\sim 1/n$ (converged). **Dictionary.** Concept i at layer ℓ read through the RMSNorm-gain linearization $(W_U \odot g)\, J_\ell$, rows normalized. **Sanity gate.** Easy single-token factual probes: 17/21 hit rank ≤20 mid-band for *both* J-lens and logit lens (ceiling). The discriminative check, multihop bridge-entity readout at the final prompt token ($n = 60$): J-lens pass@1 **0.283** vs. logit-lens 0.200 (+41% rel.); pass@20 0.547 vs. 0.586 — J-lens sharpens the rank-1 readout without expanding the top-20 shortlist, the same "replicates, less clean" flavor as Nanda's Qwen 3.6 27B replication.

2 Descriptive geometry: the workspace exists, ten times thinner

Sparse nonnegative gradient-pursuit decomposition of 200 diverse prompts (factual QA, chained arithmetic, code/SQL, prose) onto the dictionary:

- **Sparsity.** Median active concepts per (position, layer): **6** at θ=0.01, 3 at θ=0.02, $k_{90} \approx 4$, peaking at L36–42. Paper: ∼10–25.
- **Variance share.** **≤0.67%** of centered activation variance inside the band (0.91% at L60, unembedding-adjacent). Paper: 6–10%. The capacity gap (∼10×) is threshold-robust.

- **Localization.** The cleanest signature is top-1 persistence across adjacent positions: 0.19–0.20 at L32–40 vs. 0.03–0.08 at both ends — an inverted U at 50–66% depth (paper band: 33–92%).
- **Broadcast — with a catch.** Top J-directions at L24/32/40 are read (input-projection alignment $z > 3$) by **73–94** downstream components vs. **0** for matched random directions ($p \approx 10^{-24}$). But matched-dimension top *non-J* principal components are read by 77–125: wide readout is a property of high-variance directions generally, not of verbalizable ones specifically.

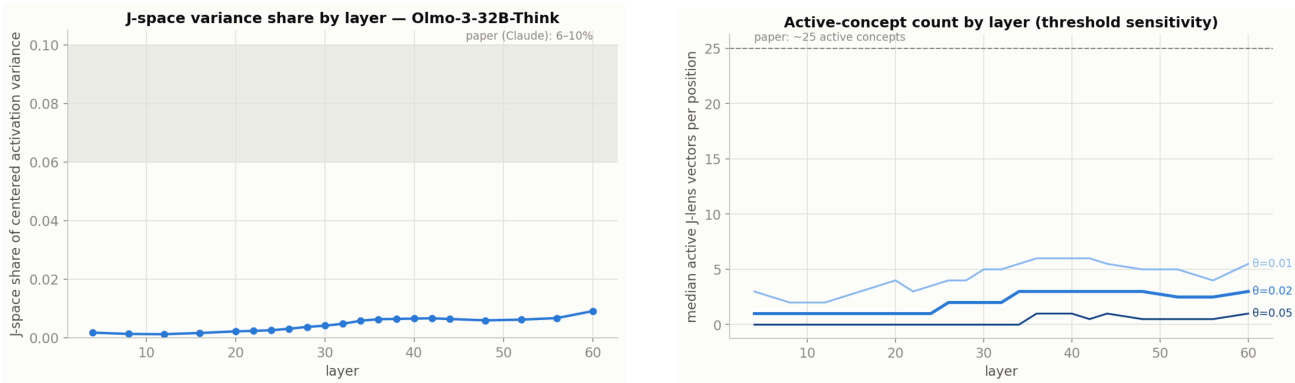

Figure 1: **Left:** J-space share of activation variance by layer; shaded band = paper's Claude range (6–10%). **Right:** median active-concept count by layer under three activation thresholds; dashed line = paper's ∼25.

3 Causal ablation: the dissociation does not replicate

Battery: 2-hop probe-swap facts (n=60), chained arithmetic (n=30), 3-table SQL with join-key checks (n=30), vs. one-hop facts, grammatical minimal pairs, held-out prose NLL. Conditions (band L20–44, doses $k \in \{10, 20, 40\}$ dims/layer, 2000-resample bootstrap CIs): static top-k J-span; matched random subspace; matched top non-J PCs; paper-faithful per-token dynamic top-10.

condition	2-hop	2-hop Δlp	arith	SQL	1-hop	grammar	NLL
none (baseline)	0.58	—	0.87	0.67	0.60	1.00	2.71
J-space dyn top-10	0.00	−23.0	0.00	0.00	0.00	0.30	24.34
J-span k=20	0.52	−0.17	0.93	0.67	0.60	1.00	2.85
random k=20	0.52	−0.08	0.90	0.67	0.60	1.00	2.75
non-J PCA k=20	0.28	**−0.86**	1.00	**0.00**	0.70	1.00	3.45

Table 1: Accuracy (and 2-hop answer-logprob delta vs. baseline) under ablation; k=10/40 follow the same pattern. Static J-span ≈ random everywhere; the *non-verbalizable* control causes the selective damage.

Three observations. (1) The paper-faithful *dynamic* condition removes whatever the model is currently computing — output degenerates to asterisks, NLL 2.71→24.34 — reproducing the "live-computation confound" Nanda flagged; it cannot count as evidence for a workspace. (2) Static J-span ablation is statistically indistinguishable from random subspaces on every task *and* on the finer answer-logprob instrument (Δ $-0.15 \ldots -0.18$ vs. $-0.01 \ldots -0.11$; CIs straddle 0). (3) The non-J high-variance control produces the only structured damage — and generation samples show its signature is *register destruction*: SQL completions lose the formal register entirely (`SELECT the user's city and the user's country, and the user's name, ...`) while arithmetic content survives. At these doses, OLMo's causal load sits in generic high-variance directions, not in the verbalizable subspace.

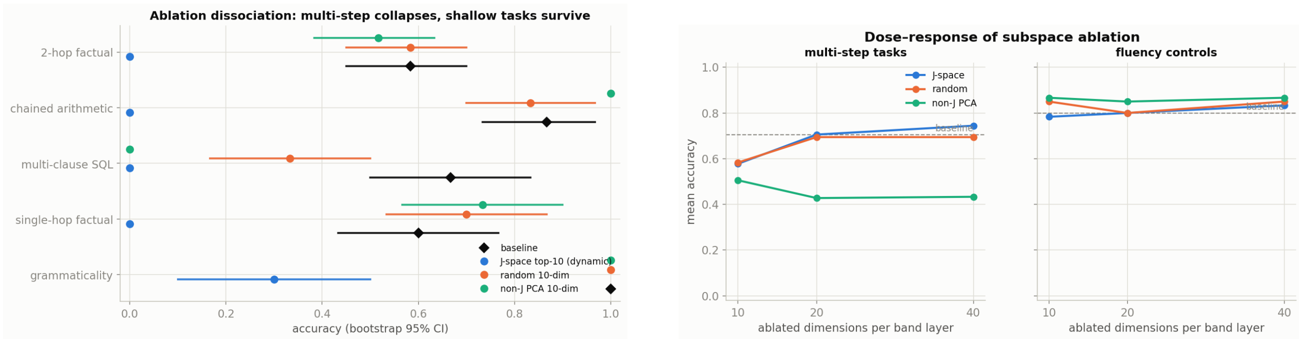

Figure 2: **Left:** per-task accuracy with 95% bootstrap CIs under the k=10 conditions. **Right:** dose–response for multi-step vs. fluency task groups.

4 The thinking-model angle: the workspace leads the chain of thought

Olmo-3-32B-Think emits explicit reasoning tokens, so the paper's identity claim (workspace content = verbalizable reasoning substrate) becomes testable in time. 90 items (40 two-hop, 30 chained arithmetic, 20 SQL join-key); J-readout at L24/32/40 at every generation step; suppressed-CoT variant closes the template's open `<think>` immediately.

- **Pre-CoT anticipation: null.** At the final prompt token the answer's median J-rank is **5613** (10% of items $\leq$20); logit lens 11610 (0%). The workspace does not precompute the answer before reasoning.
- **Intra-CoT anticipation: strong.** The answer enters a read layer's top-8 during thinking in 82–100% of items, and on traced items (n=34) the workspace holds the answer a **median 46 steps before the CoT first states it** (2-hop 48.5, SQL 38, arithmetic 9.5; workspace first in 91% of items).
- **Answer-time loading.** Forcing an immediate answer (closed `</think>`, zero reasoning tokens) pulls the answer's median rank from 5613 to **211** (23% $\leq$20), collapsing monotonically L30→L60: 2048 → 1549 → 519 → 305 → 118 → 63 → 10 → 6, with J-lens ahead of logit lens through L28–40. Suppression costs accuracy: 2-hop 0.95 → 0.82, arithmetic 1.00 → 0.90, SQL 1.00 → 1.00.
- **Faithfulness probe.** 116 events where a stable workspace top-1 word never appears in the nearby CoT (59/90 items) — but only 4 involve the final answer. The recurring divergent words are *discourse-mode* adverbs: `famously` (stable top-1 for 7–10 steps while the CoT recalls "Springfield ... birthplace of basketball"), `respectively`, `etc` during enumerations. We find no "workspace holds X, CoT argues Y, answer is X" case; the workspace appears to track the discourse mode of the current reasoning span rather than concealed content.

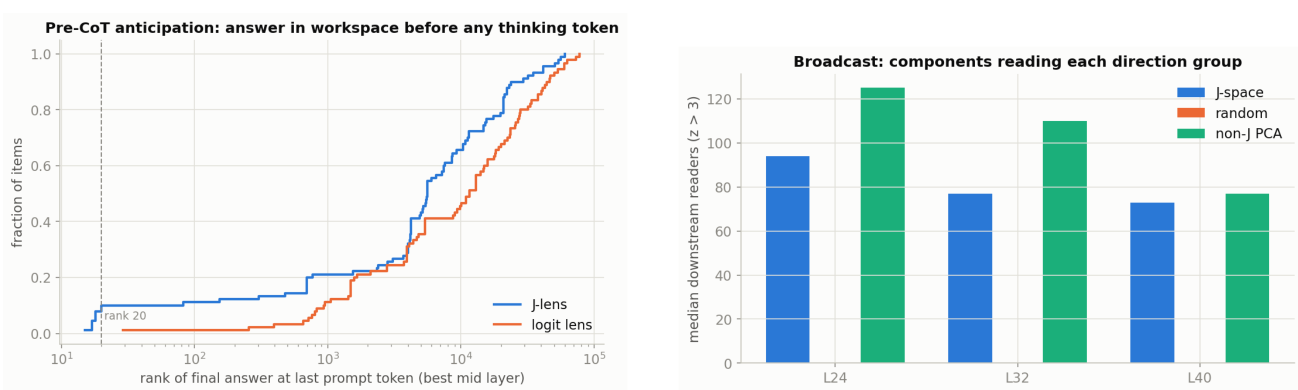

Figure 3: **Left:** ECDF of the answer's rank at the last prompt token (best mid layer), J-lens vs. logit lens — anticipation is absent in both. **Right:** median downstream readers per direction group (broadcast).

5 Stretch: an eval-awareness direction, legible but inert

Twenty tasks rendered neutrally vs. prefixed with an explicit "formal evaluation, you are being tested and scored" framing; paired J-readout differencing at the final prompt token, seven band layers. A consistent direction emerges far beyond the pre-registered criterion: sign-flip permutation $p < 10^{-3}$ at *every* layer, mean pairwise cosine of paired deltas 0.48–0.71, mean-delta norm growing with depth (22→134). From L32 upward its top decoded tokens are literally the evaluation lexicon (`evaluation`, `assessed`, `assessing`, `eval`); at L24–28 the associates are obliquer (`monitored`, `internally`, `careful`). Yet the gated mini-ablation — projecting out the mean Δh direction during generation — changed essentially nothing. Because the framing states the cue words outright, this result is best read as a clean demonstration that the J-space *verbalizes context*, not as evidence of a latent "I am being tested" state; the confound-breaking variant (implied, unstated cues) is future work.

6 Side-by-side

	Claude (paper)	Qwen 3.6 27B (Nanda)	**Olmo-3-32B-Think (ours)**
Lens recipe	1000 prompts	25 Pile prompts	120 WikiText prompts, 21 layers
Active concepts	∼10–25	not headline	6 (θ=0.01); 3 (θ=0.02)
Variance share	6–10%	—	≤0.67% (∼10× thinner)
Localization	33–92% depth	bands "less clean"	inverted U, peak 50–66% depth
Broadcast	wide readout	—	73–94 vs. 0 random; but non-J PCs 77–125
Causal dissociation	top-10 ablation kills multi-hop, spares shallow	flagged as possibly confounded	**null**: static J ≈ random; non-J PCs do the damage; dynamic = confounded collapse
Workspace vs. CoT	identity claim; CoT partially rescues	poetry/arith evals failed	pre-CoT null; **leads CoT by 46 steps**; answer-time loading under suppression

Table 2: Replication scorecard. Paper/Nanda entries as pinned in the lab's PLAN/LOG from the 2026-07-25 fetches.

7 Limitations

(1) Dictionary uses the RMSNorm-gain linearization $(W_U \odot g)J_\ell$; the final block's full nonlinearity is not inverted. (2) Fitted band 20–44 (33–70% depth) ⊂ the paper's 33–92%; the suppressed-CoT profile (rank still collapsing L44→60) suggests unfit late-band structure. (3) Active-concept thresholds are not perfectly commensurable with the paper's pursuit coefficients; the variance-share gap is the firmer capacity statement. (4) The non-J control is top-variance, not variance-*matched*; its larger effect partly reflects more energy removed — the single most important follow-up. (5) The dynamic ablation deletes live computation (Nanda's confound); a frozen prompt-selected top-k is the clean middle instrument, not yet run. (6) WikiText corpus (not Dolma); 120 fit prompts; single seed; modest n (30–60/task); CoT tasks near ceiling compress suppression contrasts. (7) Phase-5 framing states its cue words (semantic echo); its ablation was inert.

8 What to run next

(1) Variance-matched non-J controls for broadcast and ablation — energy vs. content. (2) Frozen prompt-selected top-10 ablation during generation. (3) CoT-rescue: static ablations under think-mode prompting (the paper predicts externalization rescues; our suppression asymmetry predicts it here). (4) Fit L46–62 and redo the readouts where answer formation actually happens. (5) Dolma-corpus lens — the scientific-legibility payoff of doing this on OLMo. (6) `famously`-attractor forensics across models. (7) Implied-cue eval-awareness with a behavioral endpoint.

9 The v2 delta run: resolving the open instruments

The v1 verdict left a live question: *is the causal null a property of OLMo or of our instruments?* The published Qwen comparison is narrower than "Qwen has a J-space" — its readout evals replicated while its causal interventions were flagged by the replicator himself as possibly confounded — and v1 has three known instrument gaps, each capable of producing a false null: the non-J control was not energy-matched (its damage could be an energy artifact), the static J-span is position-agnostic (could miss position-specific workspace structure), and the fitted band stops at 70% depth while v1's own suppressed-CoT profile shows answer formation continuing to L60. The v2 run closes these in order (variance matching, frozen prompt-selected ablation, late-band fit), calibrates the cot-lead headline against a false-positive floor, then tests CoT rescue. Decision tree: if the clean instruments flip the causal result, v1's null was harness; if they confirm it, the same instruments go to Qwen 3.6 27B (Neuronpedia's published lens, our harness) to separate model from method.

9.1 [v2-P4] The cot-lead headline survives its false-positive floor

The v1 claim — workspace holds the answer a median 46 steps before the CoT states it — used a permissive detector (any case/space variant first-token in any of 3 read layers' top-8, across hundreds of steps), so any word gets many chances to fire. Running the *identical* detector on matched foils (fig. 4): frequency-matched random content words fire at **0.06** (vs. **0.92** for answers) — the pure noise floor is 15× below the signal. Same-family wrong answers (other bridge entities, other join columns, nearby numbers) fire at 0.32 — but where they also appear in the CoT text their median lead is **3 steps** (vs. 46): family foils show up in the workspace *concurrently* with the text discussing them, exactly what a faithful context-verbalizing readout should do while the model enumerates candidates. Within items, the answer is detected earlier than *every* one of its foils in 66% of items (median answer detection rank 1). The 46-step lead is specific to the answer, not an artifact of detector permissiveness.

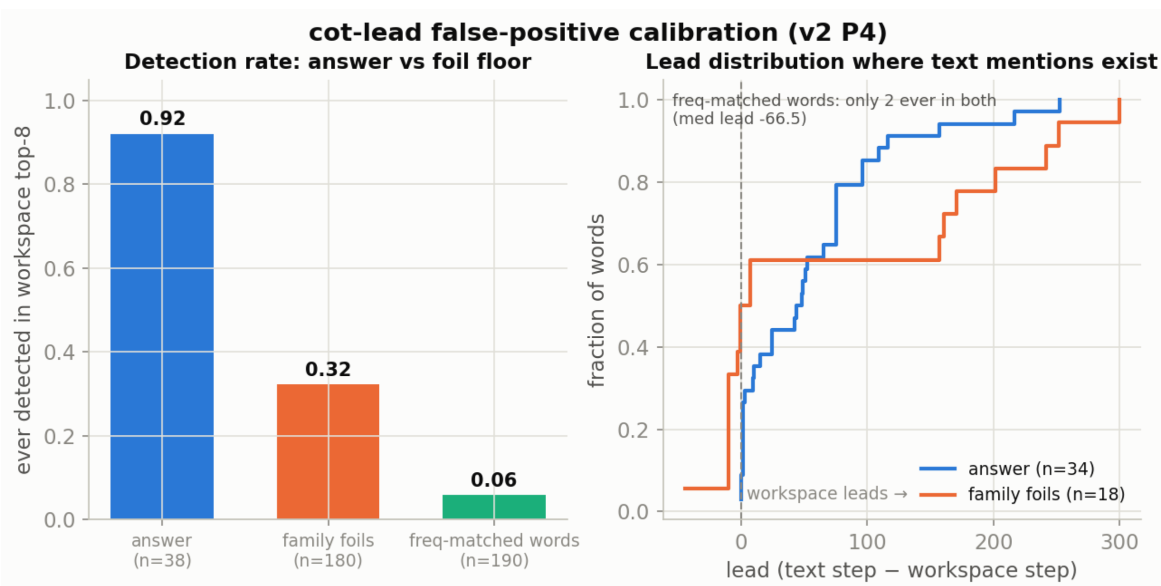

Figure 4: Foil calibration of the cot-lead detector. **Left:** detection rates — answers 0.92, family foils 0.32, frequency-matched words 0.06. **Right:** lead distributions where text mentions exist; the long positive lead is answer-specific.

9.2 [v2-P1a] v1's controls were energy-mismatched by an order of magnitude — measured, then fixed

Streaming the raw residual stream over the 200-prompt descriptive set and measuring exactly what each ablation hook removes: the k-dim J-span carries 0.5–1.8% of raw activation energy (k=10/20/40), which equals **25–33 / 44–54 / 80–94 random dimensions** — or just **1–2 / 2–5 / 3–8 deep non-J PCs**. Both of v1's rank-matched comparisons were therefore energy-confounded, in opposite directions: `random_k` removed ∼3× *less* energy than the J-span (which makes v1's "J ≈ random" *conservative* — J removed more energy and still did no more damage), while `nonJ_pca_k` (top-k PCs) removed far *more* (its damage is suspect). The v2 matched controls hit the J-span's removed energy to within 1% at every band layer and dose (fig. 5); the top PCs are so energy-fat that a contiguous *window* of deeper PCs (e.g. PCs 55–60) was needed at all 13 layers. The matched causal grid is running; its result decides whether v1's "non-J does the real damage" was content or energy.

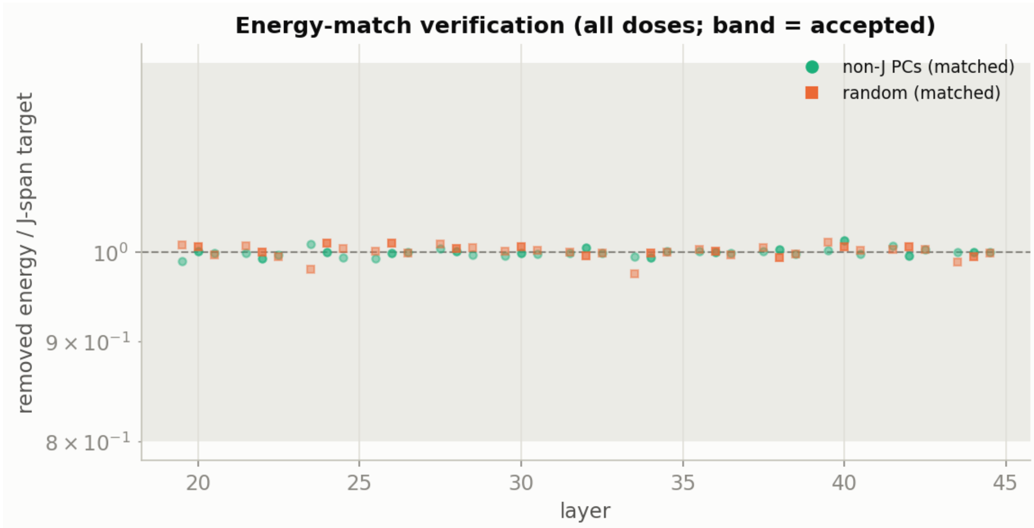

Figure 5: Energy-match verification: removed energy of each matched control relative to its J-span target (log scale), all band layers and doses. Every point sits on the 1.0 line, inside the pre-registered acceptance band.

9.3 [v2-P1b] At matched energy the causal null is clean: damage tracks energy, not content

The energy-matched grid (fig. 6) settles the question P1a raised. Every piece of v1's apparent non-J selectivity disappears: SQL returns to baseline at every dose (0.667 vs. v1's unmatched 0.000 — the "register destruction" was pure energy), the two-hop answer-logprob deltas shrink from $-0.45/-0.86/-0.93$ (CIs excluding zero) to $\mathbf{-0.01}/-\mathbf{0.17}/-\mathbf{0.15}$ (every CI straddling zero), and prose NLL sits at 2.74–2.79 against a 2.71 baseline (v1 unmatched: up to 4.02). With the confound removed in both directions, the three groups are statistically indistinguishable from each other *and from baseline* at every dose up to 40 dims/layer:

2-hop Δlogprob vs. baseline	k=10	k=20	k=40
J-span	−0.15	−0.17	−0.18
random (matched)	−0.11	−0.09	−0.08
non-J PCs (matched)	−0.01	−0.17	−0.15

Two consequences. First, the v1 causal verdict is *confirmed and strengthened*: on OLMo, statically removing the verbalizable subspace does nothing that removing an energy-equivalent arbitrary subspace doesn't do — there is no content-specific causal load at these doses, and the earlier appearance of one was an instrument artifact we have now measured and eliminated. Second, v1's own "J ≈ random" comparison was *conservative*: the J-span carries 2–5× more energy than rank-matched random dims, so v1 was comparing a heavier J-removal against a lighter control and still finding parity. The last static escape hatch for the paper's dissociation on OLMo is position-specific structure — frozen prompt-selected ablation (P2, running) and the late band (P3).

9.4 [v2-P2] The frozen instrument finds the first content-specific causal handle — and it is recall, not reasoning

The confound-free variant of the paper's intervention: per item, one clean forward pass ranks the J-dictionary by activity over the prompt, the top-10 per band layer are *frozen*, and generation proceeds with that static per-item

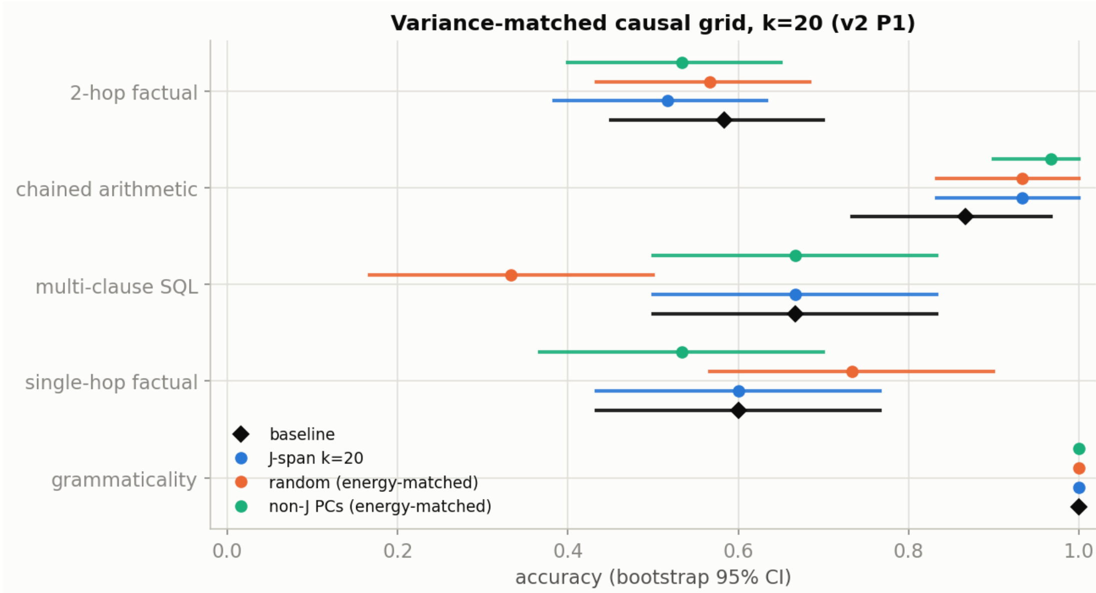

Figure 6: Energy-matched causal grid at k=20 (all doses in `v2_ablation_v2.json`). All three subspace families sit on top of each other and on baseline; contrast the v1 grid in §3, where the unmatched non-J control cratered SQL and two-hop.

projector. Two controls: the same mechanism on a random dictionary (`frozen_rand10`), and per-token live selection from a random dictionary (`live_rand10`, the matched-live control v1's dynamic condition never had).

	2-hop	2-hop lp	1-hop	arith	SQL	NLL	grammar
baseline	0.58	−1.74	0.60	0.87	0.67	2.71	1.00
frozen J-10	**0.23**	**−4.63**	**0.23**	0.87	0.67	3.00	0.85
frozen rand-10	0.53	−1.78	0.53	0.90	0.33	2.79	0.95
live J-10 (v1)	0.00	−24.8	0.00	0.00	0.00	24.3	0.30
live rand-10	0.60	−1.82	0.67	0.87	0.33	2.80	0.95

Three readings, in order of importance. **(1) A control-clean causal effect, at last.** Freezing out the item's top-10 J-directions deletes the retrieved fact: answer log-probability drops 2.9 nats (CI $[-5.49, -3.84]$ vs. baseline), recall accuracy falls 0.58→0.23 — while the identical mechanism on a random dictionary moves *nothing* (every cell within CI of baseline). Generation tasks are untouched and the stored verbatim samples under frozen-J are coherent, correct arithmetic. After P1 eliminated every subspace-level effect, the per-item selection is what matters: the J-space's causal content lives in *item-specific* directions, not in any fixed span. **(2) It is not the paper's dissociation.** Single-hop recall collapses *identically* to two-hop (both 0.23) — the paper's signature required shallow tasks to survive. What the frozen instrument found is a *factual-recall content channel*, not a multi-step scratchpad: removing the item's J-directions removes what the model knows, not how it chains. **(3) v1's live-J catastrophe was J-specific after all.** Live top-10 selection from a random dictionary does nothing (0.60 two-hop, NLL 2.80), so the v1 lobotomy was not "removing whatever aligns with the current state" generically — though the dictionaries' pool sizes differ (full vocab vs. 5120 rows), so alignment depth is a residual caveat.

Synthesis with P1: *static span removal does nothing at any matched dose; per-item frozen selection deletes facts; per-token live selection deletes the computation.* On OLMo the causal story of the J-space is content-carriage — causally real, control-clean, and increasingly precisely localized — but the workspace-as-reasoning-substrate claim remains unsupported. (The recurring SQL 0.33 cells in both random controls are the known 3-schema flaky cell; n=30 over 3 patterns.)

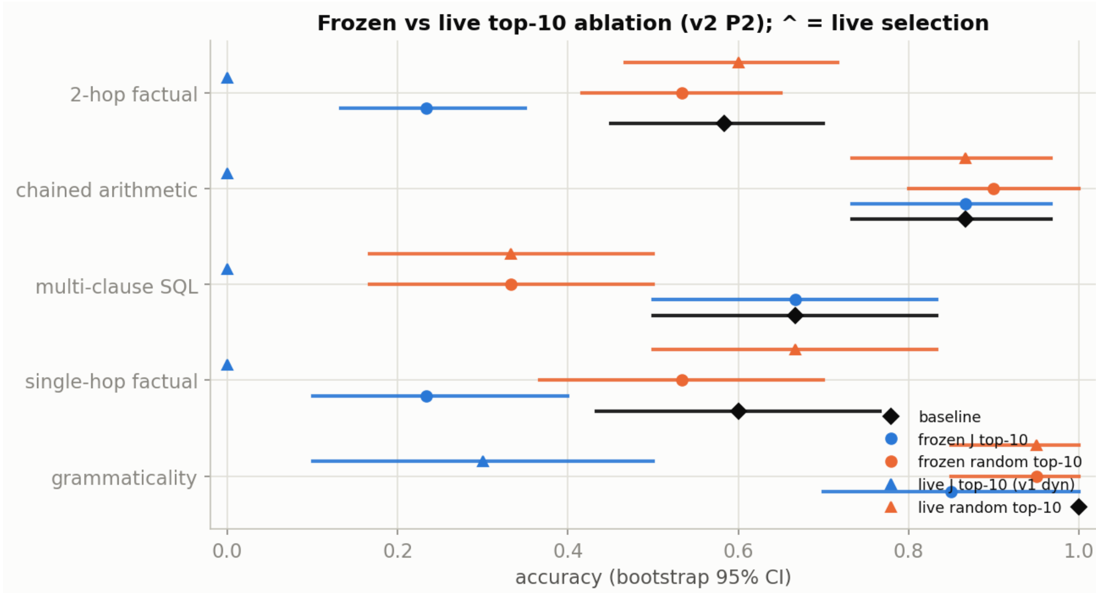

Figure 7: Frozen vs. live top-10 ablation (triangles = live selection). Frozen-J selectively destroys factual recall (1-hop and 2-hop alike) while its random-dictionary twin sits on baseline; live-J remains the uninterpretable collapse.

9.5 [v2-P3] The late band holds readout, not workspace: no late-shifted capacity, anticipation still null

v1 fitted 33–70% of depth while the paper's band reaches ∼92%, and v1's own suppressed-CoT profile showed the answer's rank still collapsing into the unfitted region — leaving open that OLMo's workspace is simply *late-shifted* and every v1 null was measured in front of it. We fitted a dedicated late-band lens on L{46,50,54,58,62} (same corpus, recipe, and seeds; backward graphs span only L46→63, so the fit cost 45 s/prompt vs. v1's 157) and redid the readouts. The fitted coverage now spans the paper's entire band.

late layer	46	50	54	58	62
variance share (%)	0.64	0.61	0.65	0.72	*1.60*
active concepts (θ=0.01)	6	5	5	4	7
top-1 persistence	0.153	0.120	0.091	0.073	0.026

(a) No late workspace. Variance share stays flat at 0.61–0.72% through L46–58 — the same ∼10×-thinner-than-Claude capacity as the mid band — and top-1 persistence *declines* monotonically (0.15→0.026), completing the inverted-U whose left side v1 measured. The one uptick, 1.6% at L62, is the unembedding-adjacent signature already seen at L60 in v1 (0.91%): it has the *lowest* persistence in the entire profile, i.e. fast-turnover next-token content, not a persistent workspace (fig. 8, left).

(b) Anticipation stays null; answer-time loading sharpens. With the dedicated late lens, the answer is still absent before thinking begins (think-mode median rank 3 916–7 291 across late layers; best-layer median 1 670, 0% of items ≤20) — refuting the "the mid-band lens just missed it" escape for the v1 pre-CoT null. Under suppressed CoT the collapse v1 glimpsed is confirmed and localized: per-layer median ranks 61→58→8→6→5 across L46–62, best-late-layer median **3** (76% of items ≤20, 58% ≤5). Notably the plain logit lens reads the late band equally well (30/18/6/4/4) — the J-lens advantage over the logit lens is a *mid-band* phenomenon; by L58–62 both lenses converge on the output distribution (final-layer median rank 4), as they must near the unembedding.

(c) The CoT lead replicates in the late band. Re-generating the 90 thinking traces (identical greedy token sequences) while reading the *late* dictionaries per step: the answer enters a late layer's top-8 a median **49.5 steps** before the CoT states it (86% of scored items, detection 99%, n=88) — statistically the same lead the mid-band dictionaries gave (46 steps, 91%), measured with an independent lens fit on disjoint layers. The lead is a property of the workspace across the full depth, not an artifact of one band's dictionary.

Consequence for the causal verdict: the last descriptive escape hatch is closed. The late band is where answers *surface*, not where a thicker workspace *lives* — so the P1/P2 causal instruments were aimed at the right band all along, and the capacity claim [SL1-C1] now covers the paper's full 33–92% depth range.

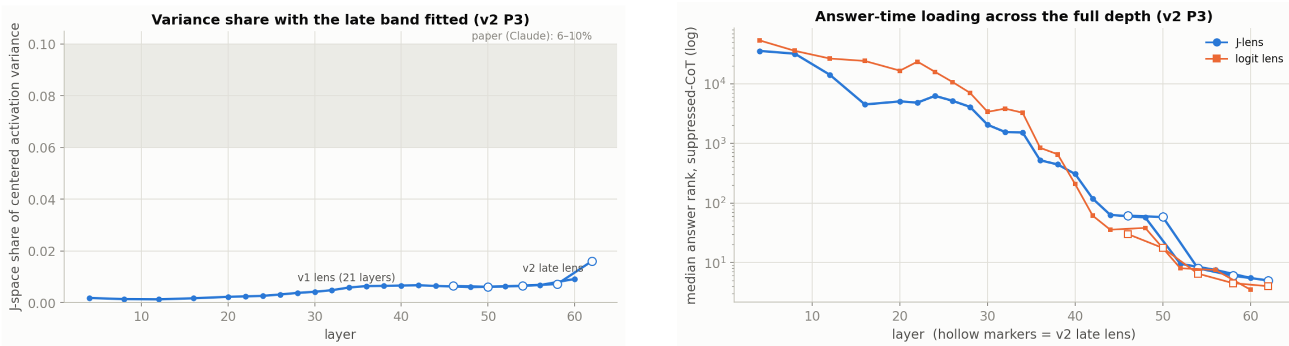

Figure 8: **Left:** variance share by layer with the v2 late lens (hollow markers) appended to the v1 profile; the paper band is shaded. **Right:** median answer rank by layer, think vs. suppressed, J-lens vs. logit lens, late band.

9.6 [v2-Q] Same instruments, second model: Qwen3.6-27B settles model-vs-harness

The standing question after P1–P3: is the causal null a fact about OLMo or about our instruments? Phase Q runs the *identical* harness on Qwen3.6-27B — the model of the published community replication — using the *published* Neuronpedia lens (1000-prompt WikiText recipe, trained by the replication's author), so any OLMo/Qwen difference cannot be blamed on our 120-prompt fit. Band = the same depth fractions, which on Qwen's 64 layers lands on the same indices (L20–44); no-think completion prompts throughout; n=30 (two-hop, one-hop), 15 (arithmetic), 20 (prose NLL).

Sanity gate. The paper's multihop bridge-entity eval, their lens under our readout: J-lens pass@1/5/20 = 0.350/0.517/0.739 vs. logit lens 0.317/0.500/0.600 (n=60) — the J-lens readout advantage replicates on Qwen, as it did on OLMo (0.283 vs. 0.200 pass@1).

Descriptive: Qwen has the paper's workspace capacity; OLMo's thinness is a model property. Same pursuit, same thresholds, same corpus:

	OLMo-3-32B (v1/P3)	Qwen3.6-27B (Q)	paper (Claude)
peak variance share	0.67% (flat 0.61–0.72)	**6.8%** (4.3→6.8 rising)	6–10%
active @ θ=0.01	≈6	32–42	10–25 (k≈25)
active @ θ=0.02	≈3	14–21	
top-1 persistence	0.19–0.20 mid-band	0.03–0.09	—

Measured by the same code path, Qwen's J-space carries ∼8–10× more of the residual stream's variance than OLMo's and holds paper-range active-concept counts. The v1 capacity anomaly [SL1-C1] is therefore a *cross-model difference*, not a harness artifact — and OLMo pairs its thin workspace with *higher* top-1 persistence, a trade-off worth follow-up. (Numerics note: Qwen's outlier-heavy activations forced two pursuit fixes — bf16 correlations and a $1/(k{+}2)$ refit step, the latter because near-duplicate dictionary atoms push the active-set Gram past the fixed-step stability bound. Stable layers moved ≤0.2pp under the fix; OLMo's banked numbers predate it and were finite throughout.)

Causal: the frozen deletion transfers; the static null transfers.

	2-hop	2-hop lp	1-hop	arith	NLL	grammar
baseline	0.87	−0.77	0.90	1.00	2.35	1.00
frozen J-10	**0.37**	**−3.19**	0.83	1.00	2.50	1.00
frozen rand-10	0.83	−0.86	0.90	1.00	2.48	1.00
J-span k20 (matched)	—	−0.85	—	—	—	—
rand k20 (matched)	—	−0.72	—	—	—	—
non-J k20 (matched)	—	−0.80	—	—	—	—

Frozen per-item J-ablation deletes the retrieved fact on Qwen exactly as on OLMo: answer log-probability -2.42 nats (cell CI $[-4.02, -2.35]$ vs. baseline $[-1.26, -0.40]$), two-hop accuracy $0.87 \to 0.37$, while the random-dictionary twin sits on baseline and generations stay coherent (verbatim frozen-J arithmetic samples are correct multi-step chains; the small NLL bump $+0.14$ appears *equally* in the random control, so it is mechanism-generic). And the energy-matched static cells are as null on Qwen as they were on OLMo: J-span, matched-random and matched-non-J all within ± 0.08 nats of baseline — on the model where the published replication reported workspace phenomena. (The static trio was also inadvertently measured once with selection rankings from the diverged pursuit pass: $-0.85/-0.77/-0.87$ — indistinguishable, i.e. the null is insensitive even to *broken* direction selection.)

One genuine cross-model asymmetry. On OLMo, frozen-J collapsed one-hop and two-hop recall *identically* (both 0.23) — pure content deletion. On Qwen, one-hop barely moves ($0.90 \to 0.83$, CIs overlap) while two-hop halves. The deletion signature on Qwen is *biased toward the composed task* — closer to the paper's shallow-tasks-survive dissociation than anything OLMo showed. Honest caveats before upgrading the claim: the one-hop items are near-ceiling capital-city probes, and a $10\times$-fatter workspace makes top-10 directions a proportionally smaller excision. Marked as the sharpest follow-up target rather than a confirmed dissociation.

Verdict on the framing question. With clean instruments the two models *agree*: static span removal does nothing, per-item frozen selection deletes retrieved content. The disagreement between the published causal story and our v1 null was *instruments all along* — while the workspace's descriptive capacity really does differ by an order of magnitude across models, and the one-hop/two-hop asymmetry hints the composed-task sensitivity may too.

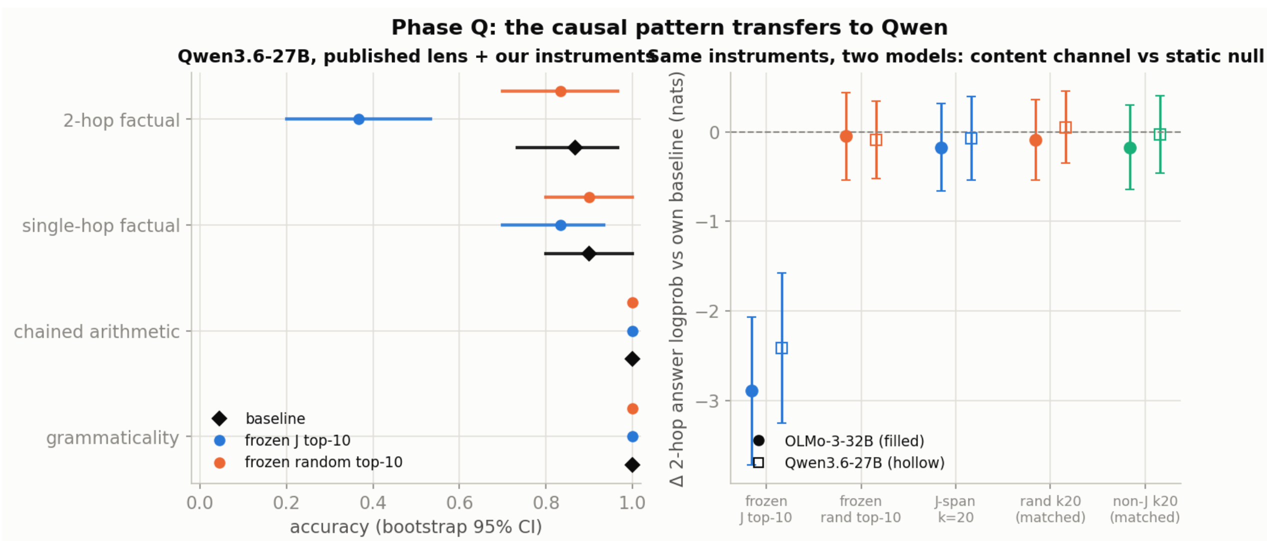

Figure 9: **Left:** Qwen causal grid — frozen-J deletes two-hop recall, controls on baseline. **Right:** Δ two-hop answer logprob vs. own baseline for both models under the same five instruments: the frozen deletion and the static null both transfer; filled = OLMo, hollow = Qwen.

Artifacts

Everything regenerates from saved metrics without model access: run dir `interpret/special-lab-1/2026-07-25_1726/` (Drive) — `report/REPORT.md` (full prose), `report/summary.json` (headline numbers), `metrics/*.json` (per-experiment), `code/special_lab1/` (pipeline s0–s10 + PLAN/LOG). v2 delta run: `2026-07-26_v2/` (metrics incl. the Qwen leg, `report/REPORT_v2.md`, `summary_v2.json`, this handout). Public draft: branch `interp_jspace` of `github.com/karlb-dev/labs` (`jspace/` + `labs/lab37_jspace_workspace.md`), PR #9. Models: `allenai/Olmo-3-32B-Think` `Qwen/Qwen3.6-27B` with lens `neuronpedia/jacobian-lens` (`qwen3.6-27b`, n=1000 WikiText). Lens engine: `github.com/anthropics/jacobian-lens` @ `581d3986`.

References

[1] Anthropic Interpretability. *Verbalizable Representations Form a Global Workspace in Language Models*. Transformer Circuits Thread, July 2026. `https://transformer-circuits.pub/2026/workspace/index.html`

[2] Anthropic. *A global workspace in language models* (research post), July 2026. `https://www.anthropic.com/research/global-workspace`

[3] Anthropic. *jacobian-lens* (reference implementation). `https://github.com/anthropics/jacobian-lens`

[4] N. Nanda. Replication review of the global-workspace results on Qwen 3.6 27B. LessWrong, July 2026.

[5] Ai2. *OLMo 3: fully open language models*, 2025–2026. `https://allenai.org/olmo`